

Chemical Tests for Common Ions

Flame Tests Method

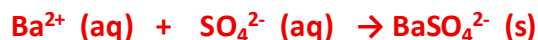
Place a damp splint into some of the ionic compound as a solid, then place the splint in a roaring Bunsen flame and look for any colour. Do not let the splint start to burn too vigorously.

Metal ion present	flame colour
Lithium, Li ⁺	red / crimson
Sodium, Na ⁺	yellow – orange
Potassium K ⁺	lilac
Calcium Ca ²⁺	brick red
Strontium Sr ²⁺	red
Barium ²⁺	apple green
Copper ²⁺	blue - green
Rubidium Rb ⁺	red / purple-red
Caesium Cs ⁺	blue - purple

Another method for flame tests involves using a nichrome wire instead of wooden splints. This can give clearer results, but you need to use reasonably concentrate acid to clean the wire.

Sulfate, SO₄²⁻

Test	Observation	Inference / conclusion
To a few cm ³ of the sulfate solution add: 2cm ³ dilute hydrochloric acid, followed by 1cm ³ of barium chloride solution	White ppt	Sulfate ions present



Carbonate, CO₃²⁻

Test	Observation	Inference / conclusion
Add dilute acid (hydrochloric or nitric) to the sample (solid or solution) then test the gas evolved by passing it through limewater in a separate test tube.	Effervescence / bubbles of gas seen. White ppt formed in limewater.	Carbonate present



Chloride, Bromide and Iodide (halide ions), X⁻

Test Cl ⁻ Br ⁻ I ⁻	Observation	Inference / conclusion
To a few cm ³ of the chloride solution add: 2cm ³ dilute nitric acid, followed by 1cm ³ of silver nitrate solution. Shake the tube carefully to mix. Add dilute aqueous ammonia to excess	White ppt. soluble in NH ₃ (aq)	Chloride ions present $\text{Ag}^+ (\text{aq}) + \text{Cl}^- (\text{aq}) \rightarrow \text{AgCl} (\text{s})$ $\text{AgCl} (\text{s}) + 2\text{NH}_3 (\text{aq}) \rightarrow [\text{Ag}(\text{NH}_3)_2]^+$
To a few cm ³ of the bromide solution add: 2cm ³ dilute nitric acid, followed by 1cm ³ of silver nitrate solution. Shake the tube carefully to mix. Add dilute aqueous ammonia then conc. aqueous ammonia to excess	cream ppt. sparingly soluble in NH ₃ (aq) Soluble in conc NH ₃	Bromide ions present $\text{Ag}^+ (\text{aq}) + \text{Br}^- (\text{aq}) \rightarrow \text{AgBr} (\text{s})$
To a few cm ³ of the iodide solution add: 2cm ³ dilute nitric acid, followed by 1cm ³ of silver nitrate solution. Shake the tube carefully to mix. Add dilute aqueous ammonia then conc. aqueous ammonia to excess	yellow ppt. insoluble in NH ₃ (aq) and conc NH ₃	Iodide ions present $\text{Ag}^+ (\text{aq}) + \text{I}^- (\text{aq}) \rightarrow \text{AgI} (\text{s})$

	Colour	Dilute NH ₃	Concentrated NH ₃
AgCl	White	Dissolves	Dissolves
AgBr	Cream	Insoluble	Dissolves
AgI	Yellow	Insoluble	Insoluble

Add Dilute NH₃ followed by Concentrated ammonia